Digital Logic Design CS201n Assignment - II

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Q1. Discuss concept ~~of~~ and significance of ECL and I^2L logic family.

Emitter coloured Logic (ECL)

ECL is a current mode logic or non saturated logic family which eleminates the turn off delay of saturated transistors by operating in the active mode. It has the fastest switching among the commercially available ICs. Typical propagation is 1ns. It requires large silicon area to dissipate high power.

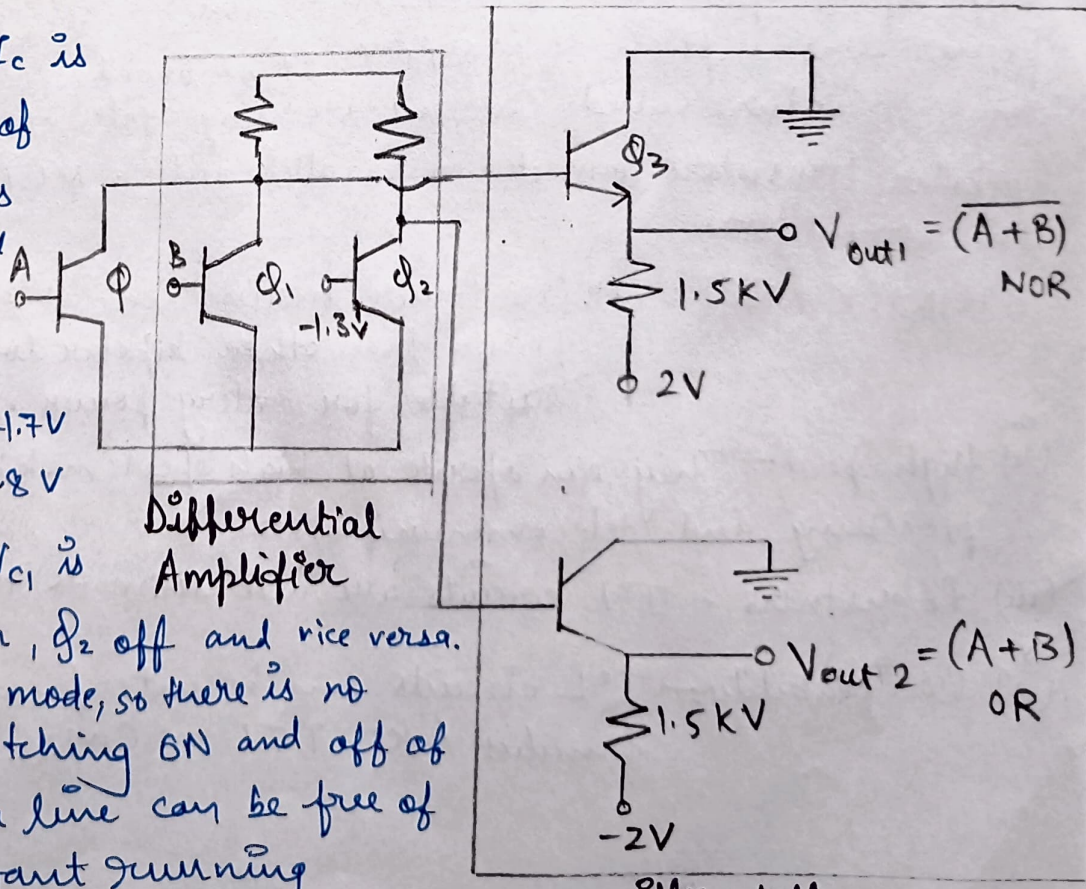
ECL OR gate or NOR gate is made of a differential amplifier cascaded with emitter follower stage. In emitter follower, output is equal to input voltage so amplification factor is 1.

Here the current I_c is constant regardless of which transister is active in the differential amplifier, depending on input logic level.

For ECL logic 0 is -1.7V and logic 1 is -0.8V

If Q_1 is ON, then V_{c_1} is low and V_{c_2} is High, Q_2 off and vice versa.

It operates in active mode, so there is no storage delay b/w switching ON and off of transistor. The power line can be free of spikes due to constant running current drawn by D.A. Emitter follower.



Emitter follower circuit is used to make the output voltage levels for logic 1 and 0 to be same as the input voltage level at B. By cascading transistors with the DA and emitter follower circuits, ECL can be used as an OR/NOR gates with multiple outputs.

SIGNIFICANCE: (i) Logic 1 = -0.8 V and Logic 0 = -1.7V

(ii) Transistor never saturate, so switching speed is high.

(iii) Noise margin is low (250 mV), so not reliable in heavy industrial environment.

(iv) It produces complementary outputs, so no need of inverters.

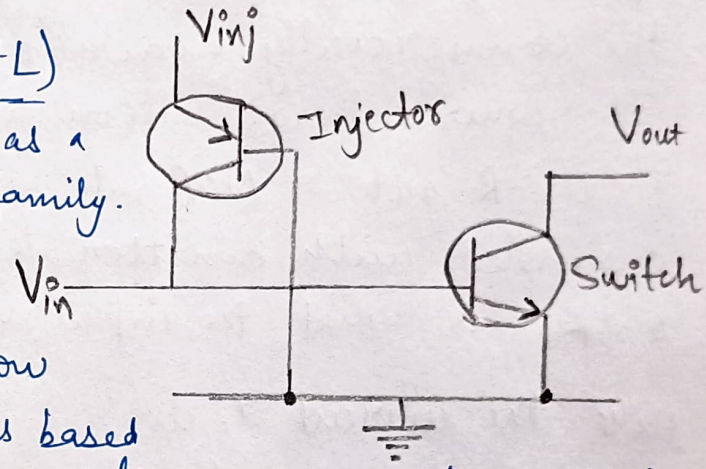
(v) Due to low output impedance, fan out is high = 25.

(vi) The typical power dissipation of ECL is 40 mW, which is higher than others.

(vii) It is suitable for very high frequency applications.

(viii) Due to -ve power supply, it is not comparable to TTL/CMOS.

INTEGRATED INJECTION LOGIC (I^2L)



I^2L is a digital logic family developed as a low power alternative to bipolar ECL family. It is a combination of BJT and MOSFET technologies which combines the high speed operation of BJT with low power consumption of MOSFET. It is based on a structure called Current Steering gate. It consists of a set of bipolar transistors connected in parallel with a MOSFET switch that controls the current flow.

SIGNIFICANCE: (i) Low Power Consumption – I^2L circuit consumes less power than other bipolar logic families, making them suitable for battery power application.

(ii) High Speed – They can operate at high speed making them suitable for data processing and telecommunication.

(iii) Robustness – I^2L circuits are insensitive to temperature and voltage fluctuation.

(iv) Compatability – I^2L circuits can be interfaced with other logic families like TTL or CMOS.

Q2. Assume an 8 bit Digital to Analog converter has a step size of 20 mV. Determine the full scale output and percentage resolution.

Given, $n = 8$ step size, $\Delta = 20\,mV$.

$\therefore$ Full scale output, $\boxed{V_{FS} = (2^n - 1) \times \Delta} = (2^8 - 1)(20 \times 10^{-3}) = 255 \times 2 \times 10^{-2}$

$\Rightarrow \boxed{V_{FS} = 5.1\,V}$

Percentage resolution, $\%R = \frac{1}{(2^n - 1)} \times 100 = \frac{100}{255} \approx 0.3921$

$\Rightarrow \boxed{\%R = 0.39\%}$

Q3. A 10 bit A/D converter of the successive approximation type has a resolution (or quantization error) of 10 mV. Determine digital output for an analogue input of 4.365 V.

We know that in successive type A/D converter, final output of its D/A converter portion always settles at a value below the analogue Input voltage to be digitized within the resolution of the converter.

Given, Analogue i/p voltage = 4.365 V

Resolution = 10 mV

$\therefore$ No. of steps $= \frac{\text{i/p voltage}}{\text{Resolution}} = \frac{4.365}{10 \times 10^{-3}} = 436.5 \approx 436$

2	436	
2	218	→ 0
2	109	→ 0
2	54	→ 1
2	27	→ 0
2	13	→ 1
2	6	→ 1
2	3	→ 0
2	1	→ 1
	0	→ 1

Thus, Step 436 will produce D/A converter output of $436 \times 10 = 4360\,mV = 4.36\,V$

and Step 437 will produce D/A converter output of $437 \times 10 = 4370\,mV = 4.37\,V$

Since it settles at step 436, digital output is Binary equivalent of 436

$(436)_{10} = (0110110100)_2$ $\boxed{\therefore \text{output} = 0110110100}$

Q4. Identify circuit shown in fig. 1 and explain its complete functioning and significance of C_1 and C_2. Also draw its all relevant output waveforms.

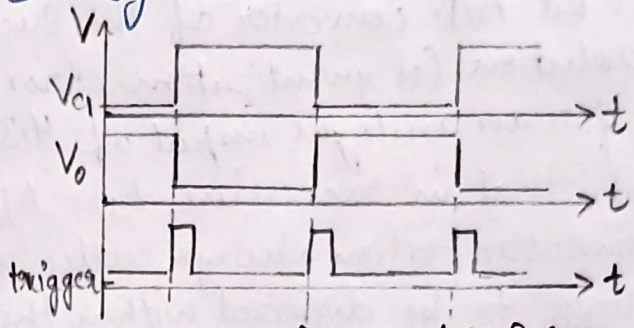
The circuit shown respresents a bistable multivibrator.

OPERATION/FUNCTIONING: Let the transistor Q_1 is ON and Q_2 is OFF at the time of switching on the circuit. (Stable state 1).

A +ve triggering pulse applied to the base of Q_2 increase its forward bias, thereby turning the transistor Q_2 ON and an increase

in collector current and a decrease in collector voltage. The fall of collector voltage is coupled to base of Q_1 where it reverse bias the base-emitter junction of Q_1. Thus Q_1 is turned off. The circuit is now on its second stable state i.e. Q_1 = OFF and Q_2 = ON and will remain in this stable state till a +ve trigger pulse is applied to base of Q_1 to switch it ON.

SIGNIFICANCE of C_1 and C_2 : C_1 and C_2 are to improve switching characteristic by passing the high frequency components of square pulses applied at A and B. They are known as commutating capacitors



OUTPUT WAVEFORMS.

Q5. Differentiate between followings: (i) RAM vs ROM

READ ONLY MEMORY (ROM)	RANDOM ACCESS MEMORY (RAM)
• It is non volatile memory i.e. content of memory will be stored when power off.	• It is volatile memory ie contents of memory will be lost if power off.
• ROM's data can't be erased or written on by user and stores essential pre-programmed information.	• It can both be written on and read from, the time it takes to access data from any random location is always the same.
• ROM can be classified as PROM, EPROM; etc.	• RAM can be classified as Static and dynamic.

(ii) SRAM vs DRAM

Static RAM	Dynamic RAM
• Stores data using a series of FFs which hold single bit in stable state. It does not require periodic refresh.	• Stores data using a capacitor and transistor and needs to be refreshed to maintain the data as capacitor discharge.
• It is faster and has higher structure density	• It is slower and have lower structure density
• It is more expensive	• It is less expensive
• It requires more power and generates more heat.	• It generic less heat and require less power
• It is used in CPU cache.	• It is used in main memory.

(iii) E^2PROM vs UVPROM

Electrically Erasable PROM	Ultraviolet Erasable PROM
• Its contents can be erased with an electrical signal without removing socket.	• Its contents can be erased by placing it under a special UV for given time, radiation discharges internal floating gate serving as program-mable connections
• Its erase time is slow.	• Its erase time is fast but we need to remove from socket
• It can be erased only a limited number of times can can be written on a limited number of times.	• It can be erased unlimited times but is only one-time programmable.
• It is more expensive and offer lower density	• It is more expensive and offer higher density.
• It is used typically for small amounts of data.	• It is used for large amount of data like firmware storage.

Q6. Realize the following logic functions using a PROM of size 8X3:

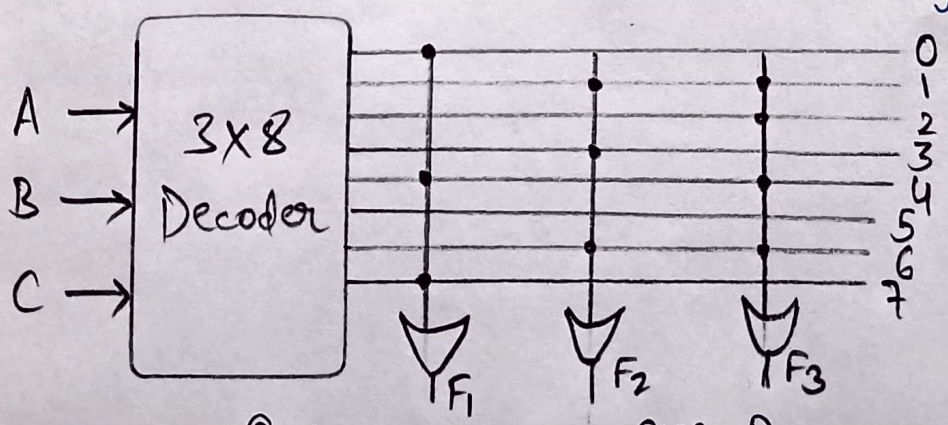
$F_1 = \Sigma m(0,4,7)$ $F_2 = \Sigma m(1,3,6)$ $F_3 = \Sigma m(1,2,4,6)$

Let us consider the truth table for the read output.

Let the inputs be A, B, C

	A	B	C	F_1	F_2	F_3
m_0	0	0	0	1	0	0
m_1	0	0	1	0	1	1
m_2	0	1	0	0	0	1
m_3	0	1	1	0	1	0
m_4	1	0	0	1	0	1
m_5	1	0	1	0	0	0
m_6	1	1	0	0	1	1
m_7	1	1	1	1	0	0

using DECODER to produce minterm



REALIZATION OF 8X3 PROM